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Physics-inspired priors for neutron star mergers improves equation of state constraints

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ABSTRACT

Gravitational-wave astronomy shows great promise in determining nuclear physics in a regime not accessible to terrestrial experiments. We introduce physics-inspired priors informed by nuclear theory and perturbative quantum chromodynamics calculations, as well as astrophysical measurements of neutron-star masses and radii. When these priors are used in gravitational-wave astrophysical inference, we show a significant improvement on nuclear equation of state constraints. Applying these to the first two observed gravitational-wave binary neutron-star mergers GW170817 and GW190425, the constraints on the radius of a $1.4\,M_\odot$ neutron star improve from $R_{1.4} = 12.54^{+1.05}_{-1.54}\,\mathrm{km}$ to $12.11^{+0.91}_{-1.11}\,\mathrm{km}$ (90% confidence intervals). LR: are these the most stringent constraints? If so, we should mention it We also show these priors can be used to perform model selection between binary neutron star and neutron star-black hole mergers; in the case of GW190425, the results are inconclusive with a Bayes factor of 1.33 in favour of the binary neutron star merger hypothesis, representing only marginal evidence. We advocate for these physics-inspired priors to be used as standard in the literature, and provide open-source code for reproducibility and further adaptation of the method.

Keywords: neutron stars — gravitational waves — equation of state

1. INTRODUCTION

One of the main goals of gravitational wave (GW) astronomy is to infer constraints on the neutron-star equation of state (EOS) from observations of binary neutron star mergers (BNS). The baryon density in neutron stars and their mergers can reach several times the nuclear saturation density $n_s = 0.16\,\mathrm{fm}^{-3}$ and temperatures of the order of tens of MeV (see, e.g., Baiotti & Rezzolla 2017; Paschalidis 2017, for reviews). Such densities and temperatures are inaccessible to terrestrial experiments and also to ab initio calculations due to the infamous fermionic sign problem, which prevents direct quantum Chromodynamics (QCD) solutions on the lattice.

A common approach to astrophysical GW inference is to use agnostic priors for some GW observables. For example, analyses of the first two observed BNS GW170817 (e.g., Abbott et al. 2017, 2018, 2019) and GW190425 (e.g., Abbott et al. 2020) employed uniform priors on the chirp mass $\mathcal{M}$ and the dimensionless tidal deformability $\tilde{\Lambda}$, implicitly assuming these parameters are uncorellated. These are the two most informative parameters for the neutron star EOS.

The above prior assumptions are overly conservative. Basic physical principles such as causality, consistency with nuclear-physics experiments, and perturbative QCD calculations, imply a strong correlation between $\mathcal{M}$ and $\tilde{\Lambda}$ in neutron-star binaries. Constraints from astrophysical measurements of neutron-star masses and radii further tightens this correlation (Altiparmak et al. 2022a; Ecker & Rezzolla 2022; Ecker & Rezzolla 2022).

Motivated by this, we introduce physics-inspired joint priors on $\mathcal{M}$ and $\tilde{\Lambda}$ that incorporate microphysical constraints from nuclear theory and perturbative QCD, as well as astrophysical constraints on neutron-star masses and radii into astrophysical GW parameter inference. We show a significant improvement on the measurement of the marginalised posterior on $\tilde{\Lambda}$ from the GW observation of GW170817 from $\tilde{\Lambda} = 466^{+901}_{-158}$ with the uniform priors to $\tilde{\Lambda} = 384^{+690}_{-226}$ with the new physics-inspired prior (confidence intervals are 90% throughout). This corresponds to an improvement in the neutron-star radius measurement of an equivalent $1.4\,M_\odot$ star from $R_{1.4} = 12.54^{+1.05}_{-1.54}$ km LR: where do these estimates come from? A reference is needed to $12.11^{+0.91}_{-1.11}\,\mathrm{km}$ LR: are these the most stringent constraints? If so, we should mention it PL: The abstract quotes the above numbers as for GW170817 and GW190425, but here is it only GW170817 – please check. For GW190425, the effective tidal deformability improves from $\tilde{\Lambda} = 381^{+1446}_{-110}$ to $\tilde{\Lambda} = 183^{+362}_{-82}$, a fractional improvement of XXX on the 90% uncertainties.

In addition to astrophysical inference, we show that these priors can be used to perform model selection, and hence distinguish between GW-source types. We perform model selection between a BNS and a neutron star – black hole (NSBH) origin for GW190425 and find a Bayes factor of 1.33 in favour of a BNS origin. When folding in the rates as the prior odds, we find an odds ratio of XXX, indicating ....

In advocating for this prior becoming standard in the literature, we provide an open-source configuration for the `Bilby` Bayesian inference library (Ashton et al. 2019; Romero-Shaw et al. 2020) currently used for the majority of GW parameter estimation by the LIGO-Virgo-KAGRA collaboration (e.g., Aasi et al. 2015; Acernese et al. 2015; Akutsu et al. 2020). This code provides a numerical implementation of the joint probability distribution as a `Bilby` constrained-prior object for $\mathcal{M}$ and $\tilde{\Lambda}$. We describe details of the method and implementation in the next Section.

2. METHODS

The construction of our physics-inspired prior closely follows that of Altiparmak et al. (2022b), which we briefly review here (see Ecker & Rezzolla 2022; Ecker & Rezzolla 2022, for additional details). We begin by building a large set of EOSs by stitching together various components. At the lowest densities ($n < 0.5\, n_s$) we adopt the Baym-Pethick-Sutherland (BPS) model (Baym et al. 1971) for the neutron-star crust. In the range $0.5\, n_s \leq n < 1.1\, n_s$, we randomly sample polytropes to span the range between the softest and stiffest EOSs from Hebeler et al. (2013). At high densities ($n \gtrsim 40\, n_s$), corresponding to a baryon chemical potential of $\mu = 2.6\,\mathrm{GeV}$, we impose the perturbative QCD constraint from Fraga et al. (2014) on the pressure $p(X, \mu)$ of cold quark matter, with the renormalization scale parameter X sampled uniformly in the range $[1, 4]$.

For the intermediate density range ($1.1\, n_s < n \lesssim 40\, n_s$), we use the parametrization method of Annala et al. (2020), which models the sound speed as a function of the chemical potential, $c_s^2(\mu)$, using piecewise-linear segments:

$$c_s^2(\mu) = \frac{(\mu_{i+1} - \mu)\, c_{s,i}^2 + (\mu - \mu_i)\, c_{s,i+1}^2}{\mu_{i+1} - \mu_i}\,, \tag{1}$$

where μ_i and $c_{s,i}^2$ are parameters defining the i-th segment in the range $\mu_i \leq \mu \leq \mu_{i+1}$. Throughout this work, we adopt natural units where $c = G = 1$, unless otherwise stated.

The number density follows as

$$n(\mu) = n_1 \exp\left(\int_{\mu_1}^{\mu} \frac{d\mu'}{\mu' c_s^2(\mu')}\right)\,, \tag{2}$$

where $n_1 = 1.1\, n_s$, and $\mu_1 = \mu(n_1)$ is set by the corresponding polytropic EOS. The pressure is then obtained via

$$p(\mu) = p_1 + \int_{\mu_1}^{\mu} d\mu'\, n(\mu')\,, \tag{3}$$

where the integration constant p_1 matches the pressure of the polytrope at $n = n_1$. We numerically integrate Eq. (3) using seven segments for $c_s^2(\mu)$.

Using this framework, we generate approximately 10^6 EOSs by randomly selecting the maximum sound speed $c_{s,\max}^2 \in [0, 1]$ and uniformly sampling the free parameters $\mu_i \in [\mu_1, \mu_{N+1}]$ (where $\mu_{N+1} = 2.6\,\mathrm{GeV}$) and $c_{s,i}^2 \in [0, c_{s,\max}^2]$. These EOSs are, by construction, consistent with nuclear theory and perturbative QCD uncertainties.

To incorporate astrophysical constraints, we solve the Tolman-Oppenheimer-Volkoff (TOV) equations for each EOS and retain only those satisfying the mass measurements of J0348+0432 (Antoniadis et al. 2013, $M = 2.01 \pm 0.04\, M_\odot$) and J0740+6620 (Cromartie et al. 2020; Fonseca et al. 2021, $M = 2.08 \pm 0.07\, M_\odot$), discarding EOSs with a maximum mass $M_{\mathrm{TOV}} < 2.0\, M_\odot$. In addition, we impose NICER radius constraints from J0740+6620 (Miller et al. 2021; Riley et al. 2021) and J0030+0451 (Riley et al. 2019; Miller et al. 2019), rejecting EOSs with $R < 10.75\,\mathrm{km}$ at $M = 2.0\, M_\odot$ or $R < 10.8\,\mathrm{km}$ at $M = 1.1\, M_\odot$. We note that, unlike what was done in Altiparmak et al. (2022a) and Ecker & Rezzolla (2022), we do not impose any GW-informed constraints on the binary tidal deformability in the construction of our prior as we are reanalysing these data using the constructed prior.

In the BNS case, the binary tidal deformability of the prior is computed as

$$\tilde{\Lambda}_{\mathrm{NSNS}} := \frac{16}{13} \frac{(12M_2 + M_1)M_1^4\Lambda_1 + (12M_1 + M_2)M_2^4\Lambda_2}{(M_1 + M_2)^5}\,, \tag{4}$$

where M_i, R_i, and $\Lambda_i = \frac{2}{3}k_2\,(R_i/M_i)^5$ are the component masses, radii, and tidal deformabilities, respectively, and k_2 is the second tidal Love number. The chirp mass is $\mathcal{M}_{\mathrm{chirp}} := (M_1 M_2)^{3/5}(M_1 + M_2)^{-1/5}$ and we define the mass ratio $q := M_2/M_1$ with $M_2 \leq M_1$.

By definition, the tidal deformability of a black hole is zero $\Lambda_{\mathrm{BH}} = 0$, implying the total tidal deformability for a NSBH binary is

$$\tilde{\Lambda}_{\mathrm{NSBH}} := \frac{16}{13} \frac{(12M_{\mathrm{BH}} + M_{\mathrm{NS}})M_{\mathrm{NS}}^4\Lambda_{\mathrm{NS}}}{(M_{\mathrm{NS}} + M_{\mathrm{BH}})^5}\,, \tag{5}$$

where M_{NS} and Λ_{NS} are the neutron-star mass and tidal deformability, respectively, and M_{BH} is the black-hole mass. The expressions for the chirp mass and mass ratio are the same, with the understanding that $M_2 = M_{\mathrm{NS}}$ and $M_1 = M_{\mathrm{BH}}$. LR: We haven't reported the analytic expressions for

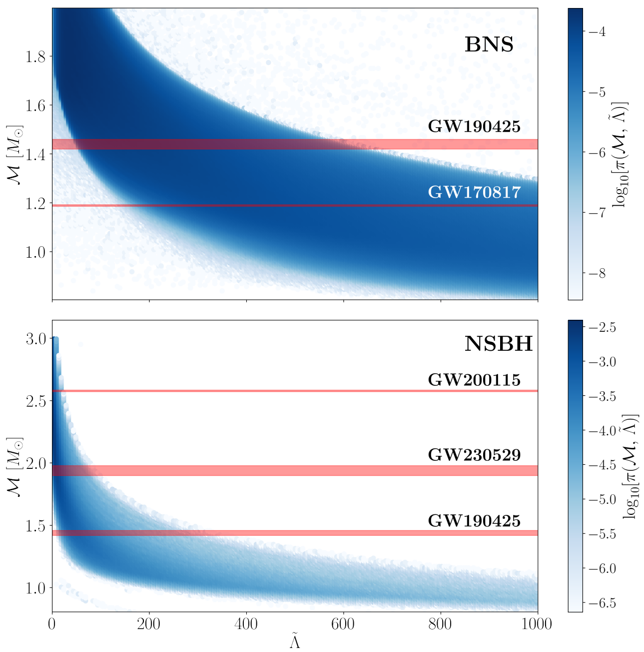

Figure 1. Coupled priors for the source-frame chirp mass $\mathcal{M}$ and dimensionless tidal deformability $\tilde{\Lambda}$. Top: prior for BNS mergers. Bottom: prior for NSBH mergers. The horizontal grey bands are the 90% credible intervals for the measured chirp masses of GW170817 (Abbott et al. 2017), GW190425 (Abbott et al. 2020), GW200115 (Abbott et al. 2021) and GW230529 (Abac et al. 2024).

the upper and lower limits; I think we should as this allows everyone to use these distributions. In the case of NSBH these would actually be new expressions CE: Indeed, we can report fitting formulas for the upper and lower bounds of 90% CI of both distributions. Note that we actually didn't report neither of these before, since we previously imposed GW constrains, which we do not do here. I'll add these fitting formulas soon. LR: ok!

2.1. *BNS and NSBH mergers*

We use the probability distributions described in the previous section as priors on the source-frame chirp mass and effective tidal deformability as shown in Fig. 1. We implement this in `Bilby` using the `conditional_prior` function, interpolating a 200×200 grid to draw from a joint prior on $\mathcal{M}$ and $\tilde{\Lambda}$.

We sample directly in source-frame chirp mass, mass ratio and the two tidal deformability parameters $\tilde{\Lambda}$ and $\delta\Lambda$ (see Favata 2014; Wade et al. 2014, for explicit expressions for these quantities). LR: what is $\delta\Lambda$? We haven't defined it. Also, later we use $\delta\tilde{\Lambda}$, which I guess is the same quantity... We note that q is not a variable that depends on the EOS, and we therefore take the standard approach of choosing a prior on q uniform in the range LR: is this range for BNS *and* NSBH? The lower limit is very small for BNS and I think it is unreasonable to expect $q \lesssim 0.3$ $q \in [0.125, 1]$ (e.g., Romero-Shaw et al. 2020)[1]. Furthermore, while both $\tilde{\Lambda}$ and $\delta\tilde{\Lambda}$ are related to the EOS, we only impose a prior here on $\tilde{\Lambda}$. In general, $\tilde{\Lambda}$ is measured much better than $\delta\tilde{\Lambda}$, which can be understood because $\tilde{\Lambda}$ enters waveform approximates at the $5^{\rm th}$ post-Newtonian order, whereas $\delta\tilde{\Lambda}$ only enters at $6^{\rm th}$ order in linear combination with $\tilde{\Lambda}$ (Wade et al. 2014). In principle, one could set a combined prior on the three parameters ($\mathcal{M}$, $\tilde{\Lambda}$, $\delta\tilde{\Lambda}$), however, adding this latter parameter would not add significantly to parameter estimation in the relatively low signal-to-noise regime. We therefore leave exploration of physically-motivated priors on $\delta\tilde{\Lambda}$ for future work.

What discussed so far refers to BNS mergers and the corresponding parameterisation does not go to large enough chirp-mass values. Hence, in the case of NSBH mergers we need to derive a new distribution by going back to the raw distributions of progenitor stellar masses and radii, and combine that progenitor distribution with that of a BH of mass M_1 and tidal deformability $\Lambda_1 = 0$. In principle, in this case we could use more sophisticated and realistic mass-ratio distributions, such as those informed by state-of-the-art binary population synthesis (e.g., Broekgaarden et al. 2021). In practice, we have decided to use here a uniformly-distributed mass-ratio prior as we have found that our results do not depend significantly on the choice made for prior distribution on mass ratios (see discussion in Appendix A). Finally, also for NSBHs we choose to work with priors in $\mathcal{M} - \tilde{\Lambda}$, enforcing $\tilde{\Lambda}(\Lambda_2, \mathcal{M}, q)$ where Λ_2 is the tidal deformability of the neutron star.

2.2. *Parameter estimation*

We perform a Bayesian parameter-estimation analysis on the GW strain data of the two (likely) BNS merger events GW170817 and GW190425 using the publicly-available data from the GWOSC repository[2]. We utilise the `Bilby` Bayesian inference framework (Ashton et al. 2019; Romero-Shaw et al. 2020) and priors on source–frame chirp mass and tidal–deformability as discussed in Section 2. Priors for all renaming parameters assume the so-called low-spin ($\chi \leq 0.05$) default priors as defined in `Bilby`. We use Planck 2018 cosmological parameters (Planck Collaboration et al. 2020) such that Hubble's constant and the matter density are respectively $H_0 = 67.66\ {\rm km\ s^{-1}}$, $\Omega_M = 0.30966$.

Parameter estimation for each run is performed using the `dynesty` nested sampler (Speagle 2020) and the `IMRPhenomPv2_NRTidal` waveform (Dietrich et al. 2019). Analysis of BNS merger signals is computationally

[1] One *could* choose a prior on q motivated by astrophysical formation scenarios (e.g.,), however we choose here to do use the common approach instead PL: Are there references for this? If not, I suggest we just delete this footnote.

[2] https://gwosc.org

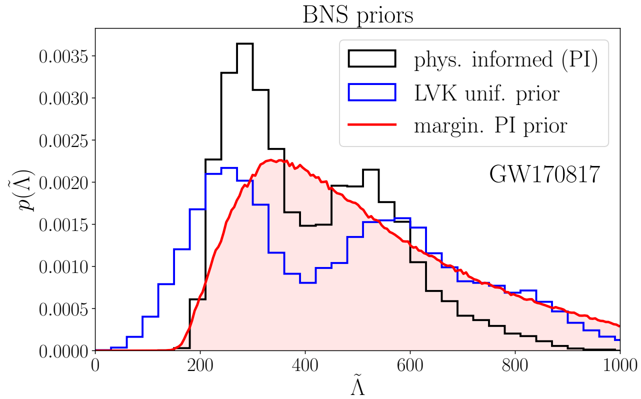

Figure 2. Marginalised posterior distributions of the tidal deformability $\tilde{\Lambda}$ obtained from Bayesian parameter estimation for GW170817. The posterior using a nuclear, microphysics, and astrophysics inspired prior is shown in teal, while the posterior using uniform priors from the LIGO-Virgo-KAGRA collaboration is shown in midnight blue. We show the (marginalised) prior with the light blue curve. Note, the posterior samples from the LVK show support up to $\tilde{\Lambda} = 2000$.

expensive and as such we utilise parallel implementation of `Bilby`, i.e., `pBilby` (Smith et al. 2020). More specifically, we reconstruct pressure–density and mass–radius curves from posterior samples of source–frame chirp mass, mass ratio, and effective tidal deformability by performing an EOS inference on the posterior samples of mass and tidal deformability. We utilise a four-piece spectral parameterisation (Lindblom 2010) stitched to an SLy (Douchin & Haensel 2001) crust at low densities (further details can be found in Appendix C).

3. RESULTS

Figure 2 shows the posterior distributions of the tidal deformability of GW170817 analyzed using the physics-inspired BNS (black), the publicly-released uniform priors from the LIGO-Virgo-KAGRA collaboration (blue; Abbott et al. 2017), and the marginalised prior for $\tilde{\Lambda}$ (red). While we show only the tidal posteriors, we sample over all binary parameters. Posterior distributions for all other binary parameters, including the chirp mass and mass ratio, are identical.

LR: up to here

Figure 3 shows the reconstructed pressure–density (top panel) and mass–radius (bottom panel) posteriors of GW170817. The shaded regions show the 90% credible intervals for the posteriors samples obtained using the physics inspired priors (teal) and uniform priors (orange). The dotted lines represent the 90% credible intervals of the effective priors on the physics prior and uniform prior when sampled with the spectral parameterisation. The grey curves show the 90% credible interval of the true EOS distribution of our priors; i.e., the distribution used to generate our chirp-mass–

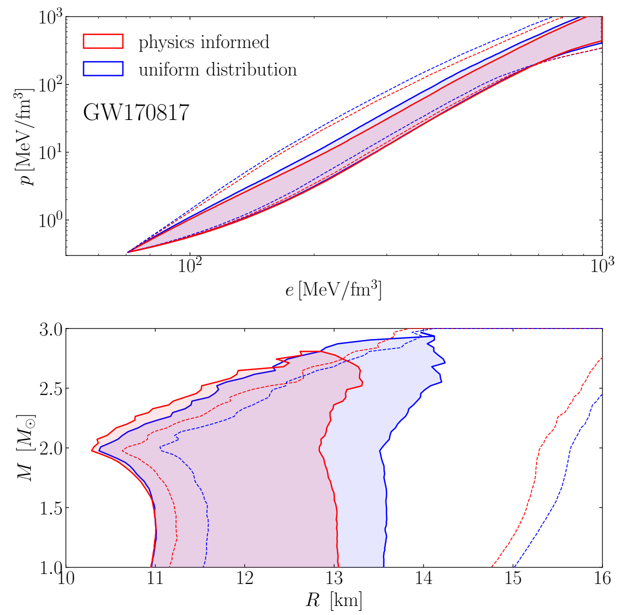

Figure 3. Posterior distributions of the cold nuclear EOS from analysis of GW170817 using the spectral parameterisation (Lindblom 2010). The top panel shows the posterior distributions in pressure–density space, while the bottom panel shows the reconstructed mass–radius curves. The shaded regions show the 90% credible intervals of the posterior distributions obtained from the analysis using the physics inspired priors (teal) and uniform priors (orange). The dashed curves show the 90% credible intervals of the effective priors, whilst the grey curve shows the 90% credible interval of the true distribution used to generate the priors. PL: bigger labels on everything. And let's discuss colour choices.CE: I suggest to show here only the comparison of the physics informed vs. uninformed posteriors. I assume the Altiparmak+ curves are the posteriors of our old paper which also includes the GW constraint. Showing them together would open a can of worms, namely that we don't use the same parametrization for prior and posterior EOS. LR: I agree; we don't need to compare with Altiparmak at least in the main text; I will revise the figures; we can make a comparison in the appendix if we fancy

tidal-deformability relationship. We find that the radius of a $1.4\,M_\odot$ neutron star can be constrained to $12.11^{+0.91}_{-1.11}$ km.

Figure 4 shows the posterior distribution of the intrinsic tidal parameter of GW190425 when analyzed using both the BNS prior (teal) and NSBH prior (pink). We show the official posterior distribution of the LVK obtained using uniform priors in blue. Bayesian-model selection between the BNS and NSBH merger gives a Bayes factor of 1.33, marginally in favour of a BNS merger origin.

Figure 5 shows the reconstructed pressure–density (top) and mass–radius (bottom) posteriors of GW190425. Equation of state the 90% credible intervals of posteriors obtained using the BNS physics inspired prior are shown in teal, while the NSBH posteriors are shown in pink. Once again, the

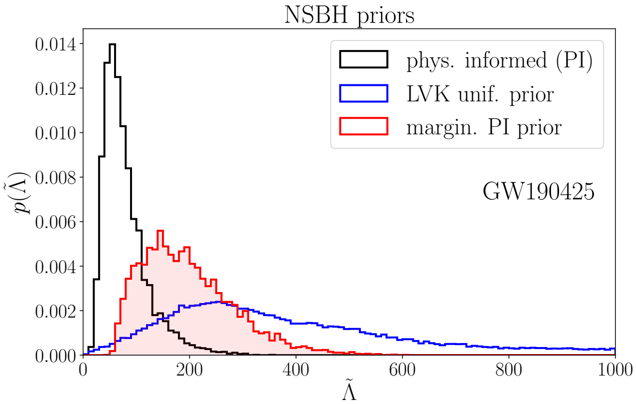

Figure 4. The posterior distributions of the tidal deformability $\tilde{\Lambda}$ obtained from Bayesian parameter estimation for G190425. The posterior using a nuclear, microphysics, and astrophysics inspired BNS prior is shown in teal, and the corresponding NSBH prior is shown in pink. The posterior using uniform priors from the LVK is shown in midnight blue.

dashed curve represents our effective prior on the EOS, while the grey curve shows the true prior. We find that the radius of $1.4\,M_\odot$ neutron star is constrained to $12.69^{+1.24}_{-1.22}$ km using the BNS prior, and $12.06^{+1.44}_{-1.15}$ km using the NSBH prior.

4. CONCLUSIONS

We have constructed new physics inspired priors for the analysis of GW strain data of BNS mergers. Analysis of the neutron star mergers GW170817 and GW190425 with our priors, constrains the effective tidal-deformability of GW170817 to $\tilde{\Lambda} = 384^{+690}_{-226}$ and GW190425 to $\tilde{\Lambda} = 183^{+362}_{-82}$. Corresponding to a 37%, and 80% decrease in the width of 90% credible interval respectively when compared to uniform priors. Performing EOS inference, we find that the radius of a $1.4M_\odot$ neutron star can be constrained to $12.11^{+0.91}_{-1.11}$ km at a 90% credible interval. Which suggests a preference for softer/stiffer equations of state ...

Model selection between a BNS and NSBH origin on GW190425, yields marginal support for a BNS origin. However, such a result is not unexpected given the low signal to noise of the event and lack of significant tidal measurements.

In this work, we have constructed priors using binaries containing neutron stars with low spins ($\chi \leq 0.05$) only. However, analysis by the LVK typically involves both galactic "low-spin" and "high-spin" ($\chi \leq 0.89$) priors. We leave the development of "high-spin" BNS and NSBH priors to future work.

APPENDIX

A. NEUTRON STAR – BLACK HOLE MERGER PRIOR GENERATION

We should detail the prior generation for NSBH here, in particular the independence of a mass ratio informed by population synthesis on the prior distributions and potentially the resultant posterior distributions. Might be worth showing an extreme example to prove the point. PL: Probably delete this appendix; make a comment in the paper, and not include Fig. 6

We generate two relations, one with uniform priors on our population of NSBH mergers, and one directly informed by state-of-the-art binary population synthesis. We use the mass ratio distribution of Broekgaarden et al. (2021) (what particular model are we using?) for the generation of our population synthesis informed priors. Figure 6 shows the distribution in mass ratio used for the generation of the prior.

B. MODEL SELECTION

We perform parameter estimation of gravitational-wave strain data using the `dynesty` nested sampler. Unlike standard MCMC samplers, `dynesty` calculates the evidence for each parameter estimation run. To perform Bayesian model selection between BNS and NSBH mergers, we calculate the Bayes factor

$$\mathrm{BF}^{\mathrm{BNS}}_{\mathrm{NSBH}} = \frac{\mathcal{Z}_{\mathrm{BNS}}}{\mathcal{Z}_{\mathrm{NSBH}}}\,, \tag{B1}$$

where $\mathcal{Z}_{\mathrm{BNS}}$ and $\mathcal{Z}_{\mathrm{NSBH}}$ are the Bayesian evidences obtained for the parameter estimation runs using the BNS and NSBH priors respectively.

Formally, for model selection we should be computing the odds ratio, which is defined as

$$\mathcal{O}^{\mathrm{BNS}}_{\mathrm{NSBH}} \equiv \frac{\mathcal{Z}_{\mathrm{BNS}}}{\mathcal{Z}_{\mathrm{NSBH}}}\frac{\pi_{\mathrm{BNS}}}{\pi_{\mathrm{NSBH}}}\,, \tag{B2}$$

where π_{BNS} and π_{NSBH} are the prior odds for each hypothesis; i.e our relative belief about which hypothesis is more likely. In this work we set the prior odds to unity, so the odds ratio is the Bayes factor. We could in principle construct an odds ratio where the prior odds was informed by the relative rates of BNS and NSBH mergers. However since these rates are still highly uncertain, we choose the more agnostic approach of setting the prior odds to unity.

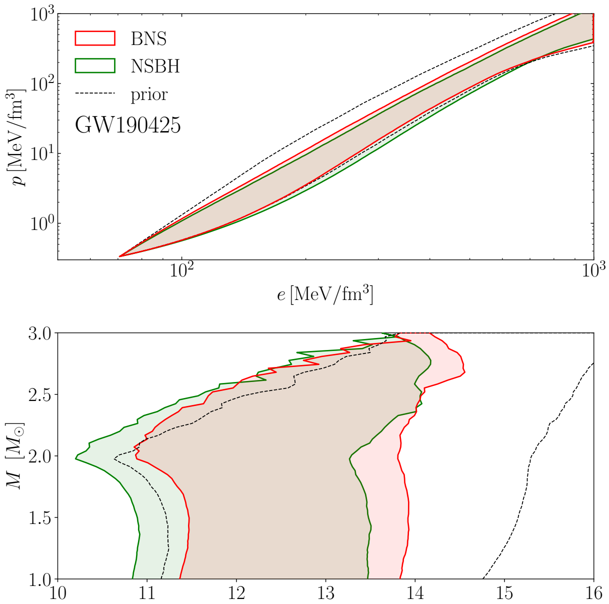

Figure 5. Posterior distributions of the cold nuclear EOS from analysis of GW190425 using the spectral parameterisation (Lindblom 2010). The top panel shows the posterior distributions in pressure–density space, while the bottom panel shows the reconstructed mass–radius curves. The shaded regions show the 90% credible intervals of the posterior distributions obtained from the analysis using the physics inspired BNS priors (teal) and the NSBH prior (orange). The black dashed curve shows the 90% credible intervals of the effective priors, whilst the grey curve shows the 90% credible interval of the true distribution used to generate the priors.

CE: Same comment as for Fig.3.

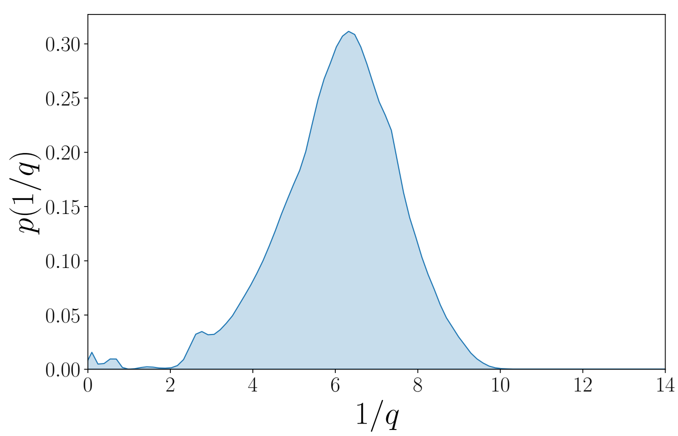

Figure 6. Probability density function of the mass ratio for detectable NSBH mergers from (Broekgaarden et al. 2021). LR: I don't think we need this figure as we can simply refer to Fig. xxx of(Broekgaarden et al. 2021)

C. EQUATION OF STATE INFERENCE

The posterior on the EOS from the GW data is

$$p(\epsilon|d) = \frac{p(d|\epsilon)\pi(\epsilon)}{\mathcal{Z}}\,, \tag{C3}$$

where $\pi(\epsilon)$ is the prior on the EOS parameters ϵ, and $\mathcal{Z}$ is the evidence. The likelihood is given by

$$p(d|\epsilon) = \int d\theta \, p(d|\theta) \, \pi(\theta|\epsilon) \, , \tag{C4}$$

where θ are our nuisance parameters which we marginalise over. Our likelihood reduces to

$$p(d|\epsilon) = \int \, d\theta \, p(d|\mathcal{M}, q, \tilde{\Lambda}) \, \pi(\mathcal{M}, q, \tilde{\Lambda}|\epsilon, \theta) \, d\mathcal{M} \, dq \, d\tilde{\Lambda} \, . \tag{C5}$$

Since we a dealing with a small number of events (i.e 1-2), we may ignore the modelling of overall neutron-star population parameters without introducing significant biases into the EOS inference (e.g Agathos et al. 2015; Wysocki et al. 2020; Landry et al. 2020). The marginalised likelihood transforms to

$$p(d|\epsilon) = \int \, p(d|\mathcal{M}, q, \tilde{\Lambda}) \, \pi(\mathcal{M}, q, \tilde{\Lambda}|\epsilon) \, d\mathcal{M} \, dq \, d\tilde{\Lambda} \, , \tag{C6}$$

We have two terms that need to be calculated; $\pi(\mathcal{M}, q, \tilde{\Lambda}|\epsilon)$ a prior on chirp mass, mass-ratio and $\tilde{\Lambda}$ for a given EOS ϵ, and $p(d|\mathcal{M}, q, \tilde{\Lambda})$ the likelihood. Typically, this likelihood would be obtained by reweighing posterior samples of parameter estimation runs

$$p(d|\mathcal{M}, q, \tilde{\Lambda}) = \frac{p(\mathcal{M}, q, \tilde{\Lambda}|d)}{\pi_{\rm PE}(\mathcal{M}, q, \tilde{\Lambda})} \, , \tag{C7}$$

however if we did this we would be removing any of our prior information from the posterior samples. Since our prior on $\tilde{\Lambda}$ and $\mathcal{M}$ is actually physical motivated, we set

$$\pi_{\rm PE}(\mathcal{M}, q, \tilde{\Lambda}) = \pi(\mathcal{M}, q, \tilde{\Lambda}|\epsilon) \, , \tag{C8}$$

which then implies that

$$p(d|\epsilon) = \int p(\mathcal{M}, q, \tilde{\Lambda}|d) \, d\mathcal{M} \, dq \, d\tilde{\Lambda} \, . \tag{C9}$$

Since the chirp mass is often very-well constrained by parameter estimation, we take the median value of chirp mass $\bar{\mathcal{M}}$, and rewrite our posterior distribution as

$$p(\mathcal{M}, q, \tilde{\Lambda}|d) = p(q, \tilde{\Lambda}) \, \delta(\mathcal{M} - \bar{\mathcal{M}}) \, , \tag{C10}$$

which simplifies the integral to

$$p(d|\epsilon) = \int p(\bar{\mathcal{M}}, q, \tilde{\Lambda}|d) \, dq \, d\tilde{\Lambda} \, . \tag{C11}$$

We note that we can also express the tidal deformability as a function of $\bar{\mathcal{M}}$, q and ϵ which removes another dimension from our integration

$$p(d|\epsilon) = \int p(q, \tilde{\Lambda}(\bar{\mathcal{M}}, q, \epsilon)|d) \, dq \, , \tag{C12}$$

which is now just an integral over mass ratio q.

We evaluate the probability density of our marginalised posterior samples using a bounded kernel-density estimate, and parameterise our EOS using the spectral parameterisation of Lindblom (2010). We use uniform priors for each EOS parameter, and enforce a minimum TOV mass $M_{\rm TOV} \geq 2.0 M_{\odot}$ and causality.

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DATA AVAILABILITY

LR: Here we can add information about the code repository on github

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